Hydrothermal Carbonization Conversion of Agricultural Residues to Value-Added Products

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Significance of Study

- **Problem:** excess **phosphorus** and **pesticides** in agricultural runoff negatively impact fisheries, drinking water, and agriculturally-important pollenating insects
- **Solution:** utilize **hydrochars** made from **inexpensive agricultural residues** to sorb phosphorus and pesticides from runoff waters

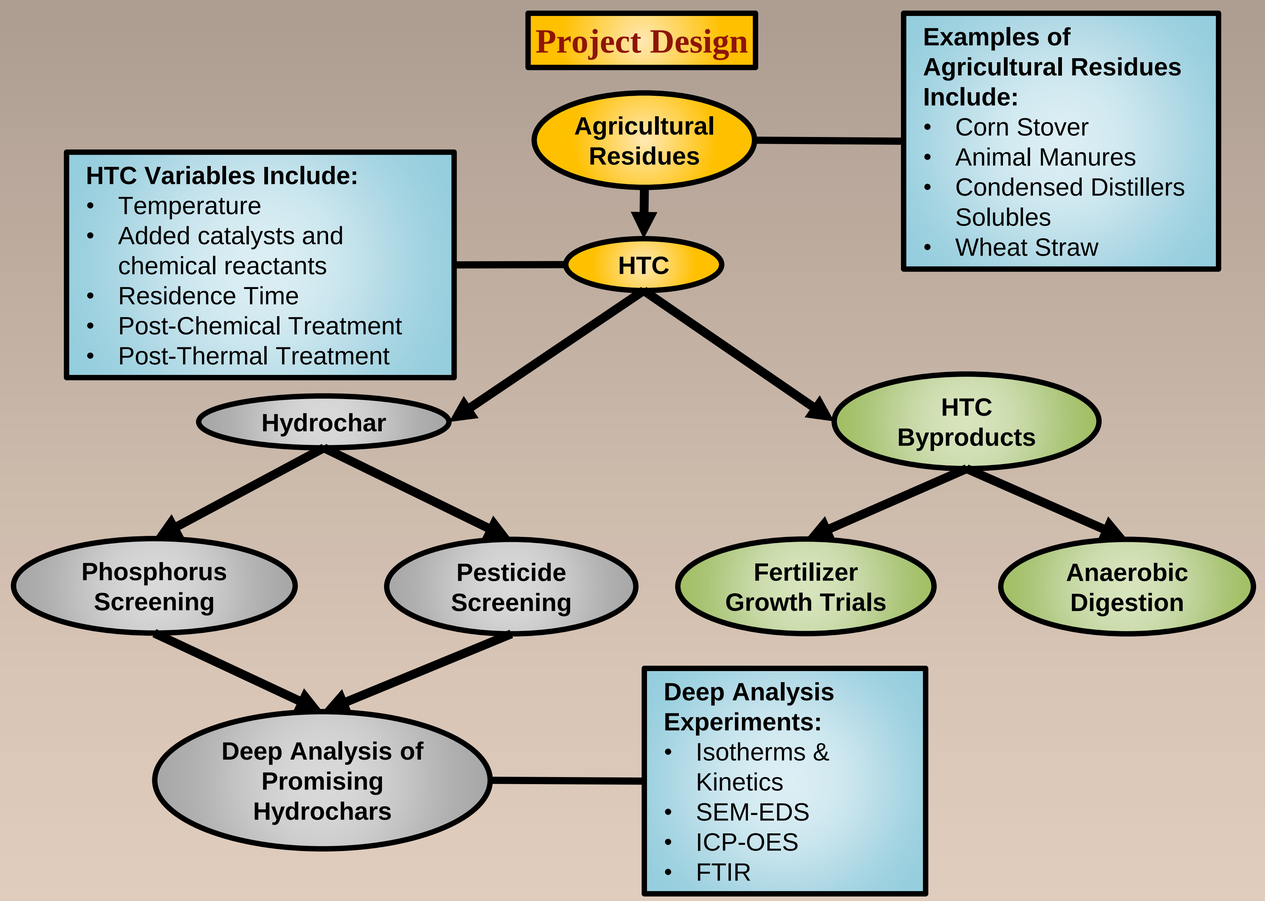


Figure 1. Effect of Phosphorus and Pesticide Pollution. Phosphorus pollution diminishes water quality and leads to ecological destabilization. Pesticide pollution directly harms wildlife.

Background

- **Carbon-based filters** (activated charcoal) are common in water filtration systems to remove chemical contaminants
- **Hydrothermal carbonization (HTC)** of organic agricultural residues produces **hydrochars** with properties similar to conventional carbon-based filters
- **HTC utilizes high heat and pressure** to convert **biomass** in aqueous solution to a hydrochar that is collected using **filtration**
- **Tailoring of hydrochars** by controlling hydrochar functionality and thermal post-treatment variables **improves** sorption and **filtering capability** of hydrochars for phosphorus and pesticides
- **Byproducts** of HTC can be utilized to **produce energy-rich biogas** or as a **crop fertilizer**

Project Design



Examples of Agricultural Residues Include:
- Corn Stover
- Animal Manures
- Condensed Distillers Solubles
- Wheat Straw

HTC Variables Include:
- Temperature
- Added catalysts and chemical reactants
- Residence Time
- Post-Chemical Treatment
- Post-Thermal Treatment

Deep Analysis Experiments:
- Isotherms & Kinetics
- SEM-EDS
- ICP-OES
- FTIR

Phosphorus Screenings

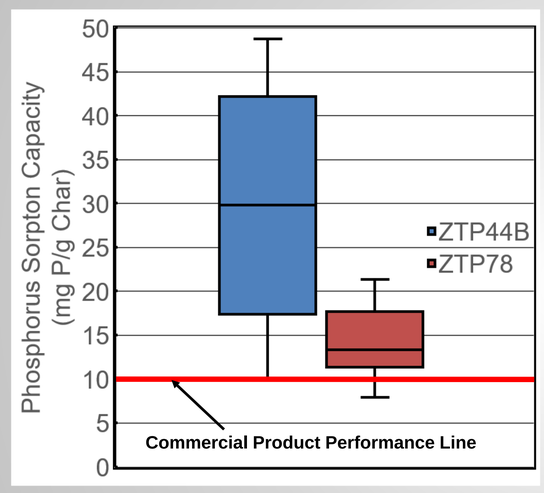


Figure 2. High Phosphorus Soprtion Chars. Chars ZTP44B and ZTP78 perform phosphorus sorption better than a commerical product.

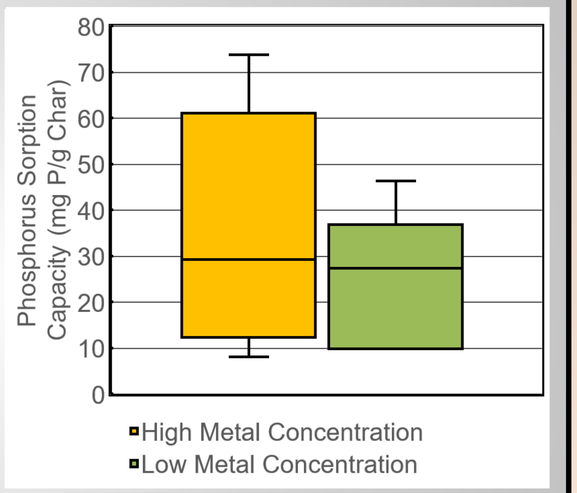


Figure 3. Effect of Added Metals. Altering the concentration of an added metal likely changes phosphorus sorption capacity.

Fertilizer Growth Trials

- Results from **2 different HTC filtrates**
- **No phytotoxic effect**
- In **fertile soils**, there was **no significant improvement** in early plant growth (expected)
- In **low fertility soils**, there was a **significant improvement** in early plant growth

Anaerobic Digestion

- HTC filtrate is **metabolizable** by **standard (non-GMO) microbes**
- Anaerobic culture mediums supplemented with HTC filtrate **produce biogas**
- Observed **biogas production is non-optimal**
- Anaerobic microbial communities may **evolve to flourish** in culture medium supplemented with HTC filtrate **resulting in optimized biogas production**

Future Directions

- **Metal Binding Mechanics:** employ leaching assays to determine **metal-hydrochar bond strength**
- **Kinetics:** discover the **rate of** phosphorus **sorption** to the hydrochar
- **Surface Chemistry:** utilize microscopy and spectrometry techniques to identify important hydrochar surface properties and chemical moeities
- **Late-Stage Growth Trials:** observe the effects of HTC filtrate fertilization on late-stage plant growth
- **Anaerobic Culture Heterogeneity:** identify prevalent microbes in anaerobic digestion cultures for biogas production

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